

Lecture Slide – 7

Naive Gaussian Elimination

Naive Gaussian Elimination

- Naive Gaussian elimination is the application of Gaussian elimination to solve systems of linear equations with the assumption that pivot values will never be zero
- The method consists of two steps:
 - **Forward Elimination:** the system is reduced to **upper triangular form**. A sequence of **elementary operations** is used.
 - **Backward Substitution:** Solve the system starting from the last variable.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22}' & a_{23}' \\ 0 & 0 & a_{33}' \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2' \\ b_3' \end{bmatrix}$$

Elementary Row Operations

- Adding a multiple of one row to another
- Multiply any row by a non-zero constant

Example

Forward Elimination

$$\begin{bmatrix} 6 & -2 & 2 & 4 \\ 12 & -8 & 6 & 10 \\ 3 & -13 & 9 & 3 \\ -6 & 4 & 1 & -18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 16 \\ 26 \\ -19 \\ -34 \end{bmatrix}$$

Part 1: Forward Elimination

Step 1: Eliminate x_1 from equations 2, 3, 4

$$\begin{bmatrix} 6 & -2 & 2 & 4 \\ 0 & -4 & 2 & 2 \\ 0 & -12 & 8 & 1 \\ 0 & 2 & 3 & -14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 16 \\ -6 \\ -27 \\ -18 \end{bmatrix}$$

Example

Forward Elimination

Step2 : Eliminate x_2 from equations 3, 4

$$\begin{bmatrix} 6 & -2 & 2 & 4 \\ 0 & -4 & 2 & 2 \\ 0 & 0 & 2 & -5 \\ 0 & 0 & 4 & -13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 16 \\ -6 \\ -9 \\ -21 \end{bmatrix}$$

Step3 : Eliminate x_3 from equation 4

$$\begin{bmatrix} 6 & -2 & 2 & 4 \\ 0 & -4 & 2 & 2 \\ 0 & 0 & 2 & -5 \\ 0 & 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 16 \\ -6 \\ -9 \\ -3 \end{bmatrix}$$

Example

Forward Elimination

Summary of the Forward Elimination:

$$\begin{bmatrix} 6 & -2 & 2 & 4 \\ 12 & -8 & 6 & 10 \\ 3 & -13 & 9 & 3 \\ -6 & 4 & 1 & -18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 16 \\ 26 \\ -19 \\ -34 \end{bmatrix} \Rightarrow \begin{bmatrix} 6 & -2 & 2 & 4 \\ 0 & -4 & 2 & 2 \\ 0 & 0 & 2 & -5 \\ 0 & 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 16 \\ -6 \\ -9 \\ -3 \end{bmatrix}$$

Example

Backward Substitution

$$\begin{bmatrix} 6 & -2 & 2 & 4 \\ 0 & -4 & 2 & 2 \\ 0 & 0 & 2 & -5 \\ 0 & 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 16 \\ -6 \\ -9 \\ -3 \end{bmatrix}$$

Solve for x_4 , then solve for x_3 ,...solve for x_1

$$x_4 = \frac{-3}{-3} = 1,$$

$$x_3 = \frac{-9 + 5}{2} = -2$$

$$x_2 = \frac{-6 - 2(-2) - 2(1)}{-4} = 1, \quad x_1 = \frac{16 + 2(1) - 2(-2) - 4(1)}{6} = 3$$

Forward Elimination

To eliminate x_1

$$\left. \begin{aligned} a_{ij} &\leftarrow a_{ij} - \left(\frac{a_{i1}}{a_{11}} \right) a_{1j} & (1 \leq j \leq n) \\ b_i &\leftarrow b_i - \left(\frac{a_{i1}}{a_{11}} \right) b_1 \end{aligned} \right\} 2 \leq i \leq n$$

To eliminate x_2

$$\left. \begin{aligned} a_{ij} &\leftarrow a_{ij} - \left(\frac{a_{i2}}{a_{22}} \right) a_{2j} & (2 \leq j \leq n) \\ b_i &\leftarrow b_i - \left(\frac{a_{i2}}{a_{22}} \right) b_2 \end{aligned} \right\} 3 \leq i \leq n$$

Forward Elimination

To eliminate x_k

$$\left. \begin{aligned} a_{ij} &\leftarrow a_{ij} - \left(\frac{a_{ik}}{a_{kk}} \right) a_{kj} & (k \leq j \leq n) \\ b_i &\leftarrow b_i - \left(\frac{a_{ik}}{a_{kk}} \right) b_k \end{aligned} \right\} k+1 \leq i \leq n$$

Continue until x_{n-1} is eliminated.

Backward Substitution

$$x_n = \frac{b_n}{a_{n,n}}$$

$$x_{n-1} = \frac{b_{n-1} - a_{n-1,n}x_n}{a_{n-1,n-1}}$$

$$x_{n-2} = \frac{b_{n-2} - a_{n-2,n}x_n - a_{n-2,n-1}x_{n-1}}{a_{n-2,n-2}}$$

$$x_i = \frac{b_i - \sum_{j=i+1}^n a_{i,j}x_j}{a_{i,i}}$$

Naive Gaussian Elimination

o The method consists of two steps

o **Forward Elimination:** the system is reduced to **upper triangular form**. A sequence of **elementary operations** is used.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \Rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22}' & a_{23}' \\ 0 & 0 & a_{33}' \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2' \\ b_3' \end{bmatrix}$$

o **Backward Substitution:** Solve the system starting from the last variable.
Solve for $x_n, x_{n-1}, \dots, x_1$.

Example 1

Solve using Naive Gaussian Elimination :

Part 1: Forward Elimination

Step 1: Eliminate x_1 from equations 2, 3

$$x_1 + 2x_2 + 3x_3 = 8$$

eq1 unchanged (pivot equation)

$$2x_1 + 3x_2 + 2x_3 = 10$$

$$eq2 \leftarrow eq2 - \left(\frac{2}{1}\right)eq1$$

$$3x_1 + x_2 + 2x_3 = 7$$

$$eq3 \leftarrow eq3 - \left(\frac{3}{1}\right)eq1$$

$$x_1 + 2x_2 + 3x_3 = 8$$

$$-x_2 - 4x_3 = -6$$

$$-5x_2 - 7x_3 = -17$$

Example 1

Part 1: Forward Elimination

$$x_1 + 2x_2 + 3x_3 = 8$$

$$-x_2 - 4x_3 = -6$$

$$-5x_2 - 7x_3 = -17$$

Step 2: Eliminate x_2 from equation 3

eq1 unchanged

eq2 unchanged (pivot equation)

$$eq3 \leftarrow eq3 - \left(\frac{-5}{-1} \right) eq2$$

$$\Rightarrow \begin{cases} x_1 + 2x_2 + 3x_3 = 8 \\ -x_2 - 4x_3 = -6 \\ 13x_3 = 13 \end{cases}$$

Example 1

Backward Substitution

$$x_3 = \frac{b_3}{a_{3,3}} = \frac{13}{13} = 1$$

$$x_2 = \frac{b_2 - a_{2,3}x_3}{a_{2,2}} = \frac{-6 + 4x_3}{-1} = 2$$

$$x_1 = \frac{b_1 - a_{1,2}x_2 - a_{1,3}x_3}{a_{1,1}} = \frac{8 - 2x_2 - 3x_3}{a_{1,1}} = 1$$

The solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Determinant

The elementary operations do not affect the determinant

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 2 \\ 3 & 1 & 2 \end{bmatrix} \xrightarrow{\text{Elementary operations}} A' = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -4 \\ 0 & 0 & 13 \end{bmatrix}$$

$$\det(A) = \det(A') = -13$$

How Many Solutions Does a System of Equations $AX=B$ Have?

Unique

$$\det(A) \neq 0$$

reduced matrix

has no zero rows

No solution

$$\det(A) = 0$$

reduced matrix

has one or more

zero rows

corresponding

B elements $\neq 0$

Infinite

$$\det(A) = 0$$

reduced matrix has

one or more zero

rows corresponding

B elements $= 0$

Examples

Unique

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} X = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

↓

$$\begin{bmatrix} 1 & 2 \\ 0 & -2 \end{bmatrix} X = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

solution :

$$X = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix}$$

No solution

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} X = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

↓

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} X = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

No solution

0 = -1 impossible!

infinte # of solutions

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} X = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

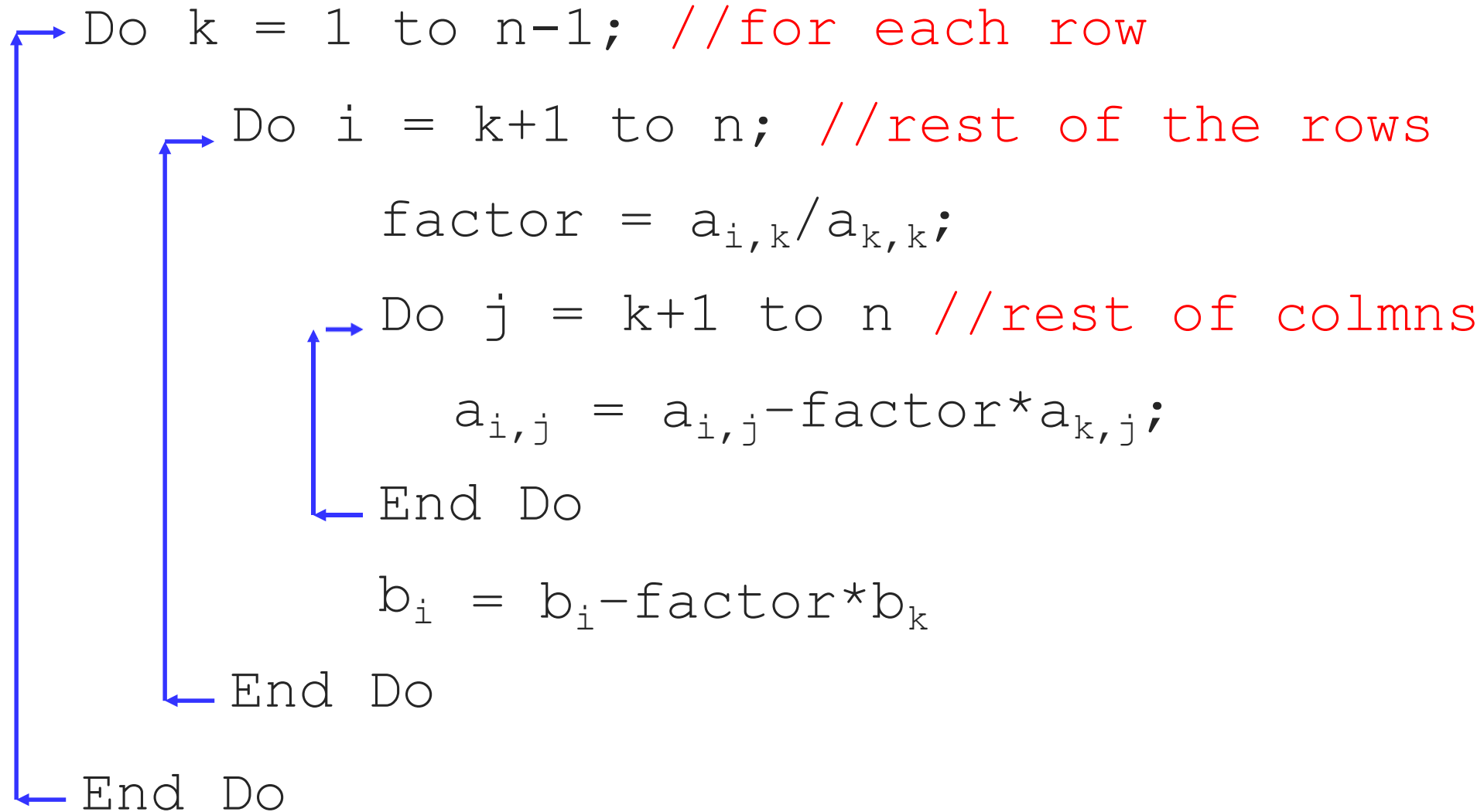
↓

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} X = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

Infinite# solutions

$$X = \begin{bmatrix} \alpha \\ 1 - .5\alpha \end{bmatrix}$$

Pseudo-Code: Forward Elimination



```
graph TD; A[Do k = 1 to n-1; //for each row] --> B[Do i = k+1 to n; //rest of the rows]; B --> C[factor = a_{i,k}/a_{k,k}]; C --> D[Do j = k+1 to n //rest of colmns]; D --> E[a_{i,j} = a_{i,j}-factor*a_{k,j}]; E --> F[End Do]; F --> G[b_i = b_i-factor*b_k]; G --> H[End Do]; H --> I[End Do];
```

Do k = 1 to n-1; //for each row

Do i = k+1 to n; //rest of the rows

factor = $a_{i,k}/a_{k,k}$;

Do j = k+1 to n //rest of colmns

$a_{i,j} = a_{i,j} - \text{factor} * a_{k,j}$;

End Do

$b_i = b_i - \text{factor} * b_k$

End Do

End Do

Pseudo-Code: Back Substitution

$x_n = b_n / a_{n,n};$ //bottom of matrix

Do $i = n-1$ downto 1; //going up

$sum = b_i;$

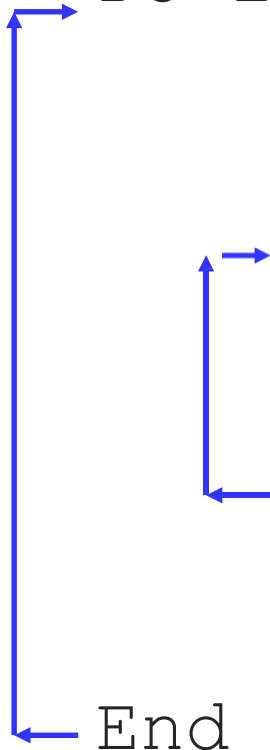
 Do $j = i+1$ to $n;$

$sum = sum - a_{i,j} * x_j;$

 End Do

$x_i = sum / a_{i,i};$

End Do

A flowchart illustrating the back substitution algorithm. It starts with the calculation of x_n . Then, a loop labeled 'Do i = n-1 downto 1; //going up' begins. Inside this loop, 'sum' is initialized to b_i . A second loop 'Do j = i+1 to n;' follows, where sum is updated by subtracting $a_{i,j} * x_j$ for each j . After the inner loop, x_i is calculated as $sum / a_{i,i}$. The flowchart uses blue arrows to show the flow from the end of the inner loop back to the start of the inner loop, and from the end of the outer loop back to the start of the outer loop.

Problems with Gaussian Elimination

- The Gaussian Elimination may fail for very simple cases.
- (The pivoting element is zero).

$$\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- Very small pivoting element may result in serious computation errors

$$\begin{bmatrix} 10^{-10} & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Example 2

Problem Statement. Use Gauss elimination to solve

$$0.0003x_1 + 3.0000x_2 = 2.0001$$

$$1.0000x_1 + 1.0000x_2 = 1.0000$$

Solution. Multiplying the first equation by $1/(0.0003)$ yields

$$x_1 + 10,000x_2 = 6667$$

which can be used to eliminate x_1 from the second equation:

$$-9999x_2 = -6666$$

which can be solved for

$$x_2 = \frac{2}{3}$$

This result can be substituted back into the first equation to evaluate x_1 :

$$x_1 = \frac{2.0001 - 3(2/3)}{0.0003}$$

Example 2

However, due to subtractive cancellation, the result is very sensitive to the number of significant figures carried in the computation:

Significant Figures	x_2	x_1	Absolute Value of Percent Relative Error for x_1
3	0.667	-3.33	1099
4	0.6667	0.0000	100
5	0.66667	0.30000	10
6	0.666667	0.330000	1
7	0.6666667	0.3330000	0.1

Note how the solution for x_1 is highly dependent on the number of significant figures.

On the other hand, if the equations are solved [in reverse order](#), the row with the larger pivot element is normalized. The equations are

$$1.0000x_1 + 1.0000x_2 = 1.0000$$

$$0.0003x_1 + 3.0000x_2 = 2.0001$$

Example 2

Elimination and substitution yield $x_2 = 2/3$. For different numbers of significant figures, x_1 can be computed from the first equation, as in

$$x_1 = \frac{1 - (2/3)}{1}$$

This case is much less sensitive to the number of significant figures in the computation:

Significant Figures	x_2	x_1	Absolute Value of Percent Relative Error for x_1
3	0.667	0.333	0.1
4	0.6667	0.3333	0.01
5	0.66667	0.33333	0.001
6	0.666667	0.333333	0.0001
7	0.6666667	0.3333333	0.00001

Thus, a pivot strategy is much more satisfactory.

Example 3

Solve the following system using Gaussian Elimination with Scaled Partial Pivoting :

$$\begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 5 & 8 & 6 & 3 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

Example 3

Initialization step

$$\begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 5 & 8 & 6 & 3 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

Scale vector:
disregard sign find
largest in magnitude in
each row

Scale vector $S = [2 \quad 4 \quad 8 \quad 5]$

Index Vector $L = [1 \quad 2 \quad 3 \quad 4]$

Why Index Vector?

- Index vectors are used because it is much easier to exchange a single index element compared to (swap) exchanging the values of a complete row.
- In practical problems with very large N , exchanging the contents of rows may not be practical.
- Scaled partial pivoting is a numerical technique in Gaussian elimination that **enhances the stability and accuracy of solving linear systems**, particularly when the coefficients in the matrix vary significantly in magnitude. By selecting the pivot element **based on its relative size within its row**, rather than its absolute magnitude, it reduces the impact of round-off errors and prevents the degradation of numerical results²³

Example 3

Forward Elimination-- Step 1: eliminate x_1

Selection of the pivot equation

$$\begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 5 & 8 & 6 & 3 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \Rightarrow \begin{cases} S = [2 & 4 & 8 & 5] \\ L = [1 & 2 & 3 & 4] \end{cases}$$

$$Ratios = \left\{ \frac{|a_{i,1}|}{|S_{l_i}|} \mid i = 1, 2, 3, 4 \right\} = \left\{ \frac{|1|}{2}, \frac{|3|}{4}, \frac{|5|}{8}, \frac{|4|}{5} \right\} \Rightarrow \max \text{ corresponds to } l_4$$

Equation 4 is the first pivot equation Exchange l_4 and l_1

$$L = [4 \ 2 \ 3 \ 1]$$

Example 3

Forward Elimination-- Step 1: eliminate x1

Update A and B

$$\begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 5 & 8 & 6 & 3 \\ \boxed{4} & \boxed{2} & \boxed{5} & \boxed{3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ \boxed{-1} \end{bmatrix}$$

First pivot equation 

$$\Rightarrow \begin{bmatrix} 0 & -1.5 & 0.75 & 0.25 \\ 0 & 0.5 & -2.75 & 1.75 \\ 0 & 5.5 & -0.25 & -0.75 \\ \boxed{4} & \boxed{2} & \boxed{5} & \boxed{3} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1.25 \\ 1.75 \\ 2.25 \\ \boxed{-1} \end{bmatrix}$$

Example 3

Forward Elimination-- Step 2: eliminate x_2

Selection of the second pivot equation

$$\begin{bmatrix} 0 & -1.5 & 0.75 & 0.25 \\ 0 & 0.5 & -2.75 & 1.75 \\ 0 & 5.5 & -0.25 & -0.75 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1.25 \\ 1.75 \\ 2.25 \\ -1 \end{bmatrix}$$

$$S = [2 \ 4 \ 8 \ 5] \quad L = [\ 4 \ 2 \ 3 \ 1]$$

$$\text{Ratios : } \left\{ \frac{|a_{l_i,2}|}{S_{l_i}} \mid i = 2,3,4 \right\} = \left\{ \frac{0.5}{4} \quad \frac{5.5}{8} \quad \frac{1.5}{2} \right\} \Rightarrow L = [\ 4 \ 1 \ 3 \ 2]$$

Example 3

Forward Elimination-- Step 3: eliminate x3

$$\begin{bmatrix} 0 & -1.5 & 0.75 & 0.25 \\ 0 & 0 & -2.5 & 1.8333 \\ 0 & 0 & 0.25 & 1.6667 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1.25 \\ 2.1667 \\ 6.8333 \\ -1 \end{bmatrix}$$

**Third pivot
equation**



$$L = \begin{bmatrix} 4 & 1 & 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1.5 & 0.75 & 0.25 \\ 0 & 0 & -2.5 & 1.8333 \\ 0 & 0 & 0 & 2 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1.25 \\ 2.1667 \\ 9 \\ -1 \end{bmatrix}$$

Example 3

Backward Substitution

$$\begin{bmatrix} 0 & -1.5 & 0.75 & 0.25 \\ 0 & 0 & -2.5 & 1.8333 \\ 0 & 0 & 0 & 2 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1.25 \\ 2.1667 \\ 9 \\ -1 \end{bmatrix} \quad L = [4 \ 1 \ 2 \ 3]$$

$$x_4 = \frac{b_3}{a_{3,4}} = \frac{9}{2} = 4.5, \quad x_3 = \frac{b_2 - a_{2,4}x_4}{a_{2,3}} = \frac{2.1667 - 1.8333x_4}{-2.5} = 2.4327$$

$$x_2 = \frac{b_1 - a_{1,4}x_4 - a_{1,3}x_3}{a_{1,2}} = \frac{1.25 - 0.25x_4 - 0.75x_3}{-1.5} = 1.1333$$

$$x_1 = \frac{b_4 - a_{4,4}x_4 - a_{4,3}x_3 - a_{4,2}x_2}{a_{l_1,1}} = \frac{-1 - 3x_4 - 5x_3 - 2x_2}{4} = -7.2333$$

Why the Scale Vector is Not Updated

Computational Efficiency: Recomputing the scale vector after every iteration would require searching all remaining elements in every row to find new maxima. This would add roughly $O(n^3)$ operations, significantly increasing the total cost of the algorithm.

Marginal Accuracy Gains: It is generally accepted that the original scale factors provide a sufficiently accurate representation of the "relative size" of each row. Numerical experiments show that the slight improvement in accuracy from re-scaling is rarely worth the high computational overhead.

Scaling before elimination is usually sufficient; dynamic scaling is not cost-effective.

Why the Scale Vector is Not Updated

Summary Table

Approach	Scale update	Cost	Accuracy gain	Used in practice?
No scaling	Never	Low	Poor for ill-conditioned rows	Rarely
Fixed scaling (once)	No	Low	Good enough	Yes
Dynamic scaling (each step)	Yes	High (×2)	Marginal	No

Example 4

Solve the following system using Gaussian Elimination with Scaled Partial Pivoting

$$\begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 5 & -8 & 6 & 3 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

Example 4

Initialization step

$$\begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 5 & -8 & 6 & 3 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

Scale vector $S = [2 \quad 4 \quad 8 \quad 5]$

Index Vector $L = [1 \quad 2 \quad 3 \quad 4]$

Example 4

Forward Elimination-- Step 1: eliminate x_1

Selection of the pivot equation

$$\begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 5 & -8 & 6 & 3 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \Rightarrow \begin{cases} S = [2 & 4 & 8 & 5] \\ L = [1 & 2 & 3 & 4] \end{cases}$$

$$Ratios = \left\{ \frac{|a_{l_i,1}|}{|S_{l_i}|} \mid i=1,2,3,4 \right\} = \left\{ \frac{|1|}{2}, \frac{|3|}{4}, \frac{|5|}{8}, \frac{|4|}{5} \right\} \Rightarrow \text{max corresponds to } l_4$$

equation 4 is the first pivot equation Exchange l_4 and l_1

$$L = [4 \ 2 \ 3 \ 1]$$

Example 4

Forward Elimination-- Step 1: eliminate x_1

Update A and B

$$\begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 3 & 1 & 4 \\ 5 & -8 & 6 & 3 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$\Rightarrow$

$$\begin{bmatrix} 0 & -1.5 & 0.75 & 0.25 \\ 0 & 0.5 & -2.75 & 1.75 \\ 0 & -10.5 & -0.25 & -0.75 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1.25 \\ 1.75 \\ 2.25 \\ -1 \end{bmatrix}$$

Example 4

Forward Elimination-- Step 2: eliminate x_2

Selection of the second pivot equation

$$\begin{bmatrix} 0 & -1.5 & 0.75 & 0.25 \\ 0 & 0.5 & -2.75 & 1.75 \\ 0 & -10.5 & -0.25 & -0.75 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1.25 \\ 1.75 \\ 2.25 \\ -1 \end{bmatrix}$$

$S = [2 \ 4 \ 8 \ 5]$ $L = [\ 4 \ 2 \ 3 \ 1]$

$$\text{Ratios : } \left\{ \left| \frac{a_{l_i,2}}{S_{l_i}} \right| \mid i = 2,3,4 \right\} = \left\{ \frac{0.5}{4} \quad \frac{10.5}{8} \quad \frac{1.5}{2} \right\} \Rightarrow L = [\ 4 \ 3 \ 2 \ 1]$$

Example 4

Forward Elimination-- Step 2: eliminate x_2

Updating A and B

$$\begin{bmatrix} 0 & -1.5 & 0.75 & 0.25 \\ 0 & 0.5 & -2.75 & 1.75 \\ 0 & -10.5 & -0.25 & -0.75 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1.25 \\ 1.75 \\ 2.25 \\ -1 \end{bmatrix}$$

$$L = \begin{bmatrix} 4 & 1 & 3 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0.7857 & 0.3571 \\ 0 & 0 & -2.7619 & 1.7143 \\ 0 & -10.5 & -0.25 & -0.75 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0.9286 \\ 1.8571 \\ 2.25 \\ -1 \end{bmatrix}$$

Example 3

Forward Elimination-- Step 3: eliminate x3

Selection of the third pivot equation

$$\begin{bmatrix} 0 & 0 & 0.7857 & 0.3571 \\ 0 & 0 & -2.7619 & 1.7143 \\ 0 & -10.5 & -0.25 & -0.75 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0.9286 \\ 1.8571 \\ 2.25 \\ -1 \end{bmatrix}$$

$$S = [2 \ 4 \ 8 \ 5] \quad L = [\ 4 \ 3 \ 2 \ 1]$$

$$\text{Ratios : } \left\{ \left| \frac{a_{l_i,3}}{S_{l_i}} \right| \mid i = 3,4 \right\} = \left\{ \frac{2.7619}{4}, \frac{0.7857}{2} \right\} \Rightarrow L = [\ 4 \ 3 \ 2 \ 1]$$

Example 4

Forward Elimination-- Step 3: eliminate x3

$$\begin{bmatrix} 0 & 0 & 0.7857 & 0.3571 \\ 0 & 0 & -2.7619 & 1.7143 \\ 0 & -10.5 & -0.25 & -0.75 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 0.9286 \\ 1.8571 \\ 2.25 \\ -1 \end{bmatrix}$$

$$L = \begin{bmatrix} 4 & 3 & 2 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 0 & 0.8448 \\ 0 & 0 & -2.7619 & 1.7143 \\ 0 & -10.5 & -0.25 & -0.75 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1.4569 \\ 1.8571 \\ 2.25 \\ -1 \end{bmatrix}$$

Example 3

Backward Substitution

$$\begin{bmatrix} 0 & 0 & 0 & 0.8448 \\ 0 & 0 & -2.7619 & 1.7143 \\ 0 & -10.5 & -0.25 & -0.75 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1.4569 \\ 1.8571 \\ 2.25 \\ -1 \end{bmatrix} \quad L = [4 \ 3 \ 2 \ 1]$$

$$x_4 = \frac{b_{l_4}}{a_{l_4,4}} = \frac{1.4569}{0.8448} = 1.7245, \quad x_3 = \frac{b_{l_3} - a_{l_3,4}x_4}{a_{l_3,3}} = \frac{1.8571 - 1.7143x_4}{-2.7619} = 0.3980$$

$$x_2 = \frac{b_{l_2} - a_{l_2,4}x_4 - a_{l_2,3}x_3}{a_{l_2,2}} = -0.3469$$

$$x_1 = \frac{b_{l_1} - a_{l_1,4}x_4 - a_{l_1,3}x_3 - a_{l_1,2}x_2}{a_{l_1,1}} = \frac{-1 - 3x_4 - 5x_3 - 2x_2}{4} = -1.8673$$

How Do We Know If a Solution is Good or Not

Given $AX=B$

X is a solution if $AX-B=0$

Compute the residual vector $R= AX-B$

Due to rounding error, R may not be zero

The solution is acceptable if $\max |r_i| \leq \varepsilon$

How Good is the Solution?

$$\begin{bmatrix} 1 & -1 & 2 & 1 \\ 3 & 2 & 1 & 4 \\ 5 & -8 & 6 & 3 \\ 4 & 2 & 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \quad \text{solution} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -1.8673 \\ -0.3469 \\ 0.3980 \\ 1.7245 \end{bmatrix}$$

$$\text{Residues : } R = \begin{bmatrix} 0.005 \\ 0.002 \\ 0.003 \\ 0.001 \end{bmatrix}$$

Remarks:

- We use index vector to avoid the need to move the rows which may not be practical for large problems.
- If we order the equation as in the last value of the index vector, we have a triangular form.
- Scale vector is formed by taking maximum in magnitude in each row.
- Scale vector does not change.
- The original matrices A and B are used in checking the residuals.